

NovaCred Financial Services Pvt. Ltd.

Digital Personal Lending & Credit Decisioning Platform

Business Requirements Document

AI-Assisted Integration Configuration & Orchestration Engine

Client	NovaCred Financial Services Pvt. Ltd.
Document Type	Business Requirements Document
Document ID	BRD-NVC-2025-004
Version	1.0 — Initial Release
Prepared By	Product & Engineering Strategy Team
Date	October 2025
Status	Final — Approved for Implementation
Classification	Confidential

Document Revision History

Version	Date	Author	Change Description
1.0	Oct 2025	Riya Desai, Head of Product	Initial draft submitted for review
1.1	Oct 2025	Karan Shah, CISO	Security and compliance addendum incorporated

1. Project Overview

NovaCred Financial Services Pvt. Ltd. is an RBI-registered NBFC headquartered in Pune, Maharashtra. NovaCred offers fully digital personal loans, salary-advance products, and buy-now-pay-later (BNPL) credit lines distributed through its mobile-first consumer application and through embedded finance partnerships with e-commerce and salary management platforms.

NovaCred intends to deploy a real-time credit decisioning engine capable of issuing personal loan approvals within 4 minutes of application submission. This requires the orchestrated integration of identity verification, credit bureau assessment, bank statement analysis, and fraud risk scoring services — all configured and managed through the AI-Assisted Integration Configuration & Orchestration Engine (hereinafter referred to as 'the Platform').

1.1 Business Context

NovaCred currently processes loan applications through a semi-manual decisioning workflow requiring 2 to 3 business days for disbursement. Applicant drop-off at the underwriting stage exceeds 55%, primarily due to document collection friction and processing delays. The proposed real-time credit engine targets

sub-5-minute approvals for eligible applicants, with same-day disbursement capability for pre-approved customers.

1.2 Scope

This document covers:

- End-to-end personal loan application and credit decisioning workflow
- Third-party API integration requirements for KYC, credit bureau, financial analysis, fraud detection, and SMS communication
- Required applicant data input fields for identity, financial profile, and device context
- Workflow orchestration including sequential and parallel verification steps and retry policies
- Authentication, credential management, and data handling requirements per RBI Digital Lending Guidelines 2022

Out of scope: loan disbursement and collections workflows, EMI payment gateway integration, and NovaCred's front-end mobile application.

2. Business Goals & Objectives

2.1 Primary Business Goals

Goal ID	Business Goal
BG-01	Reduce average personal loan approval time from 2 business days to under 4 minutes for straight-through-processing (STP) eligible applications
BG-02	Achieve a minimum STP rate of 70% for personal loan applications from applicants aged 21 to 55 with loan amounts up to INR 10 Lakhs
BG-03	Reduce identity fraud and synthetic identity applications through real-time KYC, bureau, and fraud scoring at point of application
BG-04	Comply with RBI Digital Lending Guidelines 2022, DPDP Act 2023, and UIDAI eKYC authentication norms
BG-05	Deliver real-time applicant communication via SMS at all major workflow milestones from application submission to disbursement approval

2.2 Key Performance Indicators

KPI	Target
Application-to-Approval Time (STP)	< 4 minutes
KYC Verification Success Rate	> 92%
Bureau Pull Success Rate	> 97%
Fraud Detection Coverage	100% of applications
SMS Notification Delivery Rate	> 98%
Manual Review Referral Rate	< 30%

3. Integration Requirements

NovaCred requires the following external API integrations to support the digital personal lending decisioning journey. The Platform is required to automatically parse these requirements, determine appropriate field mappings, transformation rules, and orchestration logic, and produce deployment-ready configuration artefacts.

3.1 Integration Summary

Integration	API Version	Type	Priority	Purpose
Karza KYC Provider	Latest stable	KYC / Identity	Mandatory	Aadhaar eKYC & PAN verification
TransUnion CIBIL Bureau	Latest stable	Credit Bureau	Mandatory	Credit score & bureau report
Perfios Account Aggregator	Latest stable	Financial Analysis	Optional	Bank statement & income verification
RiskGuard Analytics Engine	Latest non-beta stable	Fraud Detection	Mandatory	Real-time fraud risk scoring
Twilio SMS	Latest non-deprecated stable	Notification	Mandatory	Applicant SMS notifications

Note: NovaCred does not prescribe specific minor API versions. The Platform is expected to select the most appropriate stable, non-deprecated version for each provider based on its integration catalog, taking into account maturity, sunset timelines, and feature availability.

3.2 Mandatory Integration Requirements

3.2.1 Karza KYC Provider — Identity Verification

NovaCred requires Aadhaar-based eKYC and PAN verification for all personal loan applicants via Karza Technologies. KYC must be completed as the first step in the decisioning pipeline.

- Perform OTP-based Aadhaar eKYC using the applicant's Aadhaar-linked mobile number
- Validate PAN number and confirm Aadhaar-PAN linkage for PMLA compliance
- Execute liveness check and face match between applicant selfie and Aadhaar-linked photograph
- On KYC failure, route application to Manual Review Queue; do not auto-reject
- All KYC data must be fetched fresh per application with explicit applicant consent
- Authentication: API Key stored in secrets vault; rotation every 90 days per NovaCred security policy

3.2.2 TransUnion CIBIL Bureau — Credit Assessment

NovaCred requires individual CIBIL credit bureau reports for all personal loan applicants. Bureau pull must be triggered only after successful KYC verification.

- Pull individual credit report using the applicant's government-issued identity number and date of birth
- Retrieve credit score, active loan accounts, total outstanding balance, and delinquency history
- If score is below NovaCred's internal threshold, route to Manual Review Queue
- A no-hit result (no credit history) must not trigger automatic rejection; route to Manual Review Queue
- Retry once on HTTP 5xx or timeout before routing to Manual Review Queue
- Authentication: OAuth 2.0 Client Credentials flow; credentials stored in secrets vault

3.2.3 Perfios Account Aggregator — Financial Verification

NovaCred requires bank statement analysis via the Account Aggregator framework through Perfios. This must run in parallel with the CIBIL bureau pull after successful KYC.

- Initiate AA consent request for 6 months of bank statement data from the applicant's nominated bank
- Retrieve average monthly inflow, salary credits, existing EMI obligations, and cash irregularity indicators
- If applicant declines AA consent or data is unavailable, fall back to manual income document review
- Poll for consent completion up to 3 times at 2-minute intervals before routing to Manual Financial Review Queue
- Authentication: OAuth 2.0 Client Credentials; AA consent tokens are applicant-scoped and session-bound

3.2.4 RiskGuard Analytics Engine — Fraud Risk Scoring

NovaCred mandates real-time fraud scoring on all applications. RiskGuard must run in parallel with KYC at application initiation and must not block the KYC step.

- Submit applicant identity reference, mobile number, originating IP, device session hash, and requested loan principal to RiskGuard
- Receive fraud risk score and risk band classification
- Score ≥ 75 : Immediately suspend processing; escalate to Fraud Investigation Team
- Score 50–74: Apply Enhanced Due Diligence (EDD) flag; continue with mandatory manual review at final decision
- Score 25–49: Apply watchlist flag; continue automated processing
- Score < 25 : Continue normal STP processing
- If RiskGuard is unavailable after one retry, apply EDD flag and continue; must not block pipeline
- Authentication: Bearer token stored in secrets vault

3.2.5 Twilio SMS — Applicant Communication

NovaCred requires transactional SMS notifications via Twilio at defined milestones. All SMS must use DLT-registered Sender ID 'NCRDLP'. SMS failure must not block any workflow step.

- Send SMS on application submission with reference number
- Send SMS on successful KYC completion
- Send SMS with AA consent link when Perfios consent request is generated
- Send SMS on loan approval with approved amount and next steps
- Send SMS if application is referred to manual review with expected turnaround
- Send SMS if application is suspended due to fraud risk escalation
- Retry 2 times on delivery failure at 3-minute intervals; log non-delivery for compliance audit
- Authentication: Account SID and Auth Token stored in secrets vault; all calls over HTTPS

4. Applicant Data Input Requirements

The following fields are collected from the personal loan applicant during the digital application journey. The Platform is expected to automatically determine how each field maps to the corresponding integration API parameters. Field names below reflect NovaCred's internal applicant data model and may differ from vendor API parameter names.

4.1 Identity & Contact Fields

Field Name	Format / Validation	Required	Description
applicant_full_name	String, as per Aadhaar	Yes	Legal name for KYC matching and bureau pull
applicant_phone	10-digit Indian mobile	Yes	Aadhaar-linked mobile for eKYC OTP and SMS notifications
applicant_email_id	Valid email format	Yes	For loan agreement delivery and digital communication
dob_dd_mm_yyyy	DD/MM/YYYY	Yes	Date of birth; used for bureau pull and age eligibility check
pan_id	AAAAA9999A format	Yes	Permanent Account Number for bureau pull, KYC, and AML compliance
aadhaar_uid	12-digit numeric	Yes	Required for Karza eKYC; to be masked in all logs and storage
home_address_text	Multi-line address string	Yes	Current residential address for bureau and risk zone mapping
address_pincode	6-digit numeric	Yes	For serviceability check and regional risk classification
kyc_consent_flag	Boolean (true/false)	Yes	Explicit consent for Aadhaar eKYC; must be captured before Karza API is called
bureau_consent_flag	Boolean (true/false)	Yes	Explicit consent for CIBIL enquiry under CICRA 2005
aa_data_consent_flag	Boolean (true/false)	Yes	Explicit consent for Account Aggregator financial data sharing
request_ip_address	IPv4 or IPv6	Yes	Originating IP captured at submission; submitted to RiskGuard
session_device_hash	System-generated string	No	Device fingerprint submitted to RiskGuard if available

4.2 Financial Profile Fields

Field Name	Format / Validation	Required	Description
work_type	Salaried / Self-Employed / Business	Yes	Determines income verification pathway and Perfios analysis type
monthly_take_home	Numeric, INR	Yes	Self-declared monthly net income; cross-verified against Perfios bank analysis

loan_ask_amount	Numeric, INR, min 10,000 max 10,00,000	Yes	Requested loan principal; submitted to RiskGuard for fraud scoring
repayment_months	Numeric, 6 to 60	Yes	Desired loan tenure in months
savings_account_num	9 to 18 digit numeric	Yes	Primary bank account for Perfios AA financial data linkage and EMI auto-debit
bank_branch_ifsc	11-character alphanumeric	Yes	Must correspond to a valid RBI-registered bank branch
active_loan_count	Numeric, 0 to 10	Yes	Self-declared number of active loans; cross-verified with CIBIL data
employer_name	String	No	Employer or business name; used for risk zone assessment
preferred_bank_for_aa	String (bank name)	No	Applicant's preferred bank for AA consent initiation; defaults to all linked FIPs if not provided

4.3 Integration-Specific Input Fields

4.3.1 Perfios Account Aggregator Fields

Field Name	Format / Validation	Required	Description
customer_id	Alphanumeric string, non-empty	Yes	Unique applicant identifier issued by Perfios AA; used to scope consent request and retrieve financial data for the correct customer session
date_range_from	ISO 8601 date, YYYY-MM-DD; must be in the past	Yes	Start date for the bank statement data fetch window; must cover at least 6 months prior to the application date as required by the AA consent framework
date_range_to	ISO 8601 date, YYYY-MM-DD; must not be after application date	Yes	End date for the bank statement data fetch window; typically set to the current date at time of application submission
consent_artifact	Alphanumeric string, non-empty	Yes	required

4.3.2 Twilio SMS Fields

Field Name	Format / Validation	Required	Description
from	String; DLT-registered Sender ID, max 11 characters	Yes	Twilio Sender ID for outbound SMS; must be set to the DLT-registered value 'NCRDLP'

			per TRAI regulations for transactional messaging
body	String; max 160 characters per SMS segment; UTF-8	Yes	Templated SMS message content for the applicable workflow milestone; must use DLT-approved message templates only
account_sid	String; begins with 'AC', 34-character alphanumeric	Yes	Twilio Account SID used to authenticate API requests; must be retrieved from the secrets vault and must never be hardcoded or logged in plain text
to	String; begins with 'AC', 34-character alphanumeric	Yes	Twilio Account SID used to authenticate API requests; must be retrieved from the secrets vault and must never be hardcoded or logged in plain text

4.3.3 RiskGuard Analytics Engine Fields

Field Name	Format / Validation	Required	Description
applicant_id	Alphanumeric string, non-empty; unique per application	Yes	NovaCred-internal unique reference for the loan applicant; submitted to RiskGuard to correlate the fraud risk score with the correct application record

5. Credit Decisioning Workflow

The following describes the end-to-end applicant journey for digital personal loan decisioning. The workflow is designed for straight-through processing where all verifications pass automated thresholds.

Step	Activity
Step 1	Applicant submits digital loan application with all required fields from Section 4
Step 2	System validates input field format and completeness before invoking any external API
Step 3	RiskGuard fraud scoring initiated in parallel with Karza KYC verification
Step 4	Karza KYC verification — Aadhaar eKYC, PAN verification, liveness and face match
Step 5	Perfios AA consent request and CIBIL bureau pull initiated in parallel after successful KYC
Step 6	Perfios financial analysis and CIBIL report retrieval completed
Step 7	Automated credit decisioning engine evaluates all verification outputs and risk scores
Step 8	Loan approval notification via SMS; loan agreement generated

6. Retry & Fallback Requirements

Integration	Retry Policy & Fallback Behaviour
Karza (KYC)	No retry for eKYC OTP step — single attempt per session. On failure: route to Manual Review Queue. Fresh consent required for any re-attempt.
CIBIL (Bureau)	Retry 1 time on HTTP 5xx or timeout. On second failure: route to Manual Review Queue. No-hit must not trigger rejection.
Perfios (AA)	Poll for AA consent up to 3 times at 2-minute intervals. If incomplete: route to Manual Financial Review Queue.
RiskGuard (Fraud)	Retry once after 30-second interval. On second failure: apply EDD flag and continue. Must not block pipeline.
Twilio (SMS)	Retry 2 times on delivery failure at 3-minute intervals. Non-delivery must not block any workflow step.

7. Authentication & Security Requirements

Integration	Required Authentication Method
Karza KYC	API Key stored in secrets vault; rotation every 90 days; passed as request header
CIBIL Bureau	OAuth 2.0 Client Credentials flow; credentials stored in secrets vault
Perfios AA	OAuth 2.0 Client Credentials for Perfios API; AA consent tokens are applicant-scoped and session-bound
RiskGuard	Bearer token stored in secrets vault; EDD fallback if authentication fails
Twilio SMS	Account SID and Auth Token stored in secrets vault; all calls over HTTPS only

General requirements: All credentials must be stored in a designated secrets vault and never hardcoded. Aadhaar numbers must be masked in all logs. PAN numbers must be masked in logs. All inter-service communication must use TLS 1.2 or higher.

8. Hook & Lifecycle Requirements

The Platform is required to automatically configure the appropriate integration lifecycle hooks based on the requirements in this document.

8.1 Pre-Call Hooks

- Credential resolution must occur immediately before every API call; credentials fetched from vault and not cached beyond a single request
- Field-level encryption must be applied to Aadhaar numbers, PAN numbers, and bank account numbers before inclusion in outbound payloads
- Rate limit guard checks must execute before every call to CIBIL and Perfios; calls must be queued if rate limit utilisation reaches 85%
- Pre-authentication token validation must confirm valid credentials before each call; expired tokens must trigger re-authentication

8.2 Post-Call Hooks

- All API responses must be validated against expected response schema before downstream processing
- Post-call transformations must be applied immediately after successful response validation
- All API calls and responses (with PII masked) must be emitted to the audit trail upon completion
- All API calls must be logged to NovaCred's observability platform with request ID, integration name, response time, HTTP status, and outcome

8.3 Failure Hooks

- Automated retry must be triggered per the retry policies in Section 6
- On terminal failure, the on-failure alert hook must notify NovaCred's integration operations team within 2 minutes
- All retry attempts must be individually logged in the audit trail

9. Operational & Compliance Requirements

9.1 Manual Review Queues

Queue Name	Trigger Condition
Manual Review Queue — KYC	Karza eKYC fails; face match score below threshold; PAN-Aadhaar mismatch
Manual Review Queue — Credit	CIBIL no-hit; score below threshold; API failure after retry
Manual Financial Review Queue	Perfios AA consent not completed; data unavailable from FIP
Fraud Investigation Queue	RiskGuard score ≥ 75 ; all automated processing suspended

9.2 Data Handling & Compliance

- All personal data must be stored within India per RBI Digital Lending Guidelines 2022 and DPDP Act 2023
- Applicant consent for each data processing activity must be recorded with timestamp and IP address before the corresponding API is invoked
- Financial data from Perfios AA must be purged within 24 hours of loan decision as per AA framework data minimisation requirements
- CIBIL credit data must be retained for a maximum of 7 years per RBI CICRA guidelines
- SMS notification logs must be retained for 2 years for TRAI compliance